

Let, $x \in \emptyset$. This is a contradiction. This can happen when both $x \in A$ and $x \notin A$.
 By Definition 5, $x \in A$. By Defn. of AND, $x \in A \wedge x \notin A$. By Defn. 2, $x \in A \cap \bar{A}$.
 By Defn. 3 of Section 2.1, $\boxed{\emptyset \subseteq A \cap \bar{A}}$

$$\therefore A \cap \bar{A} = \emptyset \quad \square$$

10) Show that a) $A - \emptyset = A$ b) $\emptyset - A = \emptyset$

Solution: a) Let, x be an arbitrary element of $A - \emptyset$, i.e. $x \in A - \emptyset$.
 By Defn. 4, $x \in A \wedge x \notin \emptyset$. By Defn. of AND, $x \in A$ and $x \notin \emptyset$.
 By Defn. 3 from Section 2.1, $\boxed{A - \emptyset \subseteq A}$.

Let, x be an arbitrary element of A , i.e. $x \in A$.
 • By Defn. of empty set, $x \notin \emptyset$. By Defn. of AND, $\boxed{x \in A \wedge x \notin \emptyset}$.
 • By Defn. 4, $x \in A - \emptyset$.
 • By Defn. 3 from Section 2.1, $\boxed{A \subseteq A - \emptyset}$. $\therefore A - \emptyset = A \quad \square$

b) Let, x be an arbitrary element of $\emptyset - A$, i.e. $x \in \emptyset - A$.
 By Defn. 4, $x \in \emptyset \wedge x \notin A$. By Defn. of AND, $x \in \emptyset$ and $x \notin A$.
 By Defn. 3 from Section 2.1, $\boxed{\emptyset - A \subseteq \emptyset}$.

Let, x be an arbitrary element of $\emptyset$, i.e. $x \in \emptyset$. By Defn. of empty set, there does not exist any such x . $\therefore \boxed{x \in \emptyset \wedge x \notin A}$ must be True by vacuous ness. By Defn. 4, $x \in \emptyset - A$.
 By Defn. 3 from Section 2.1, $\boxed{\emptyset \subseteq \emptyset - A}$. $\therefore \emptyset - A = \emptyset \quad \square$

11) Let A and B be sets. Prove the commutative laws from Table 1 by showing that a) $A \cup B = B \cup A$ b) $A \cap B = B \cap A$

Solution: a) Let x be an arbitrary element of $A \cup B$, i.e. $x \in A \cup B$.
 By Defn. 1, $x \in A \vee x \in B$. By 1st Commutative Law of propositions, $x \in B \vee x \in A$. By Defn. 1, $x \in B \cup A$. By Defn. 3 from Section 2.1, $\boxed{A \cup B \subseteq B \cup A}$.